

NABEEL AHMED KHAN

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SUMMARY

Senior Backend Architect (6+ YOY) specializing in designing regulated, enterprise-grade financial infrastructure and high-throughput exchange systems. Expert in **Node.js** and **NestJS**, focusing on event-driven microservices, distributed transaction processing, and digital asset custody. Proven track record of bridging Web2 banking rails with Web3 protocols, ensuring strict compliance, security, and zero-financial-drift reconciliation for high-volume platforms.

TECHNICAL SKILLS

Backend Architecture: NestJS, TypeScript, Node.js, Microservices, Event-Driven Architecture, gRPC, WebSockets
Data & Infrastructure: PostgreSQL (Optimization), Redis, MongoDB, AWS (KMS, SES, Lambda), Docker, Kubernetes
Systems & Security: Kafka/RabbitMQ, Distributed Systems, Idempotency, JWT/OAuth2, RBAC, CI/CD, Shamirs Secret Sharing
Fintech & Web3: Exchange Architecture (CEX/DEX), Custody, Smart Contract Integration (EVM/Solana)

EXPERIENCE

Senior Software Engineer & Technical Lead

Mar 2023 – Present

BlockApex

Karachi, PK (Hybrid)

- Architected **BullionFX**, a regulated Commodities Exchange backend using **NestJS**. Designed a "Dual-Ledger" system syncing high-speed SQL order books with Stellar blockchain settlements for instant trade execution.
- Implemented institutional-grade custody solutions using **Shamirs Secret Sharing** and **AWS KMS**, splitting private keys across isolated environments to meet strict compliance and security standards.
- Built the backend for **Midl**, a P2P crypto-fiat bridge handling **10,000+ daily users**. Engineered an **Event-Sourced** architecture using **BullMQ** to decouple transaction ingestion from processing, ensuring resilience in traffic spikes.
- Implemented strict **idempotency keys** and distributed locking strategies (Redis) to prevent double-spending and ensure zero financial drift between off-chain databases and on-chain settlements.
- Developed **Elektrik**, a DEX with an $O(\log n)$ in-memory Order Matching Engine capable of processing high-frequency limit orders via WebSockets, optimizing for sub-millisecond latency.
- Led the migration of monolithic services to a **Microservices** architecture, improving deployment velocity by **40%** via automated CI/CD pipelines.

Full Stack Blockchain Engineer

May 2022 – Mar 2023

Xord (Acquired by BlockApex)

Karachi, PK

- Solely architected **Script TV**, a high-throughput "Watch-to-Earn" platform. Built a custom indexing layer in **Node.js** that synced on-chain rewards with off-chain video analytics in real-time.
- Implemented secure wallet integration flows using **JWT-based authentication** linked to cryptographic signatures (EIP-712), ensuring secure user onboarding for enterprise clients.
- Optimized complex PostgreSQL queries for a blockchain explorer, achieving a **40% reduction in API latency** for historical data retrieval.

Co-Founder & CTO

Dec 2020 – May 2022

Tezeract

Karachi, PK

- Led and mentored a development team of **4 engineers** in building 5+ AI-centric SaaS products, securing early-stage funding.
- Designed backend infrastructure for high-availability video processing pipelines deployed in **Dubai Mall** kiosks, utilizing edge devices.
- Integrated Sabre APIs for an Online Travel Agency (OTA) application, creating a payment reconciliation engine that processed **\$50k+ in simulated monthly transaction volume**.

Lead MERN Stack Engineer

May 2020 – Dec 2020

Xehen

Karachi, PK

- Managed a **3-person team**, establishing REST API standards and database schemas that supported the platform's initial **5,000 active users**.

EDUCATION

Virtual University of Pakistan

Master of Science in Computer Science

Lahore, PK

2024 – 2026

University of Karachi (UBIT)

Bachelor of Science in Software Engineering

Karachi, PK

2017 – 2021